

INFO53883 - AI & ML Capstone Project

PERFORMANCE, QUALITY, AND ETHICAL CONSIDERATION
REPORT (D3 PREPARATION) – 10%

Due: Week 12 (softcopy on SLATE, hardcopy to your professor in class)

Name 1: _____ Student Number 1: _____

Name 2: _____ Student Number 2: _____

Name 3: _____ Student Number 3: _____

Name 4: _____ Student Number 4: _____

Name 5: _____ Student Number 5: _____

Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your professor in class in Week 12.

This report is **not the Final Technical Report**.

Its purpose is to evaluate the current performance of your AI/ML solution and demonstrate that your team has critically assessed the quality, reliability, and ethical implications of your project.

Your **Project Proposal** and **Project Design Document** have already covered:

- Project planning
- Work Breakdown Structure (WBS)
- Gantt Chart
- Git/GitHub workflow
- Risk management
- Project milestones
- Expected outcomes
- AI/ML methodology selection
- System architecture

Do not repeat this information in this report.

Instead, focus on evaluating the implementation and the results achieved so far.

This report requires your team to critically evaluate the **performance, quality, and ethical considerations** of your AI/ML solution. You must present measurable evidence demonstrating how well your system performs using appropriate evaluation metrics, document your testing and quality assurance activities, analyze the current limitations of your solution, and reflect on your development process.

This report must be supported by quantitative evidence. Simply stating that your model performs well is not sufficient. Performance claims must be justified using mathematical calculations and interpretation.

Report Structure

1. Current Project Progress (0.5 mark)

Provide a brief overview of your project's current implementation status.

Include:

- Current project completion percentage
- Features successfully implemented
- AI/ML model(s) completed
- Remaining work before the Final Technical Report

This section should be approximately **half a page**.

2. Performance Evaluation and Quality Assessment (3 Marks)

Evaluate the current performance of your AI/ML solution using evaluation metrics that are appropriate for your project.

Examples include (but are not limited to):

Classification

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Regression

- MAE
- MSE
- RMSE
- R^2 Score

Clustering

- Silhouette Score

Natural Language Processing (NLP)

- Accuracy
- Precision
- Recall
- F1 Score
- BLEU Score (where applicable)

Your discussion should include:

- Appropriate evaluation metrics
- Performance tables
- Graphs or charts
- Confusion Matrix (where applicable)
- Sample predictions or outputs
- A clear explanation of what each metric indicates about your model's performance

You must also describe the testing and quality assurance activities completed during development, such as:

- Model validation and testing methods
- Dataset verification and preprocessing
- Error analysis
- Quality assurance or quality control procedures
- Actions taken to improve model performance and reliability

Your discussion should demonstrate both the **performance** and the **quality** of your solution.

3. Mathematical Justification of Model Performance (4 Marks)

This section is mandatory.

Marks will be awarded based on your ability to mathematically justify your model's performance.

For **every evaluation metric** presented in Section 2, you must include:

- The mathematical formula
- The values used in the calculation
- The final calculated result
- A detailed explanation of what the result means for your project

Examples include:

- Accuracy
- Precision
- Recall
- F1 Score
- RMSE
- MAE
- R^2 Score
- Any additional metric appropriate for your project

Important

Simply inserting formulas, screenshots, or values is **not sufficient**.

You must clearly explain:

- Why the metric was selected
- What the calculated value indicates
- Whether the result demonstrates good, acceptable, or poor model performance
- Any limitations identified from the mathematical results

Performance claims without mathematical justification will not receive marks.

Reports that only include screenshots of software outputs or evaluation scores without explaining the mathematics behind the results will receive little or no credit for this section.

4. Ethical Considerations and System Limitations (2 Marks)

Evaluate your AI/ML solution from both an ethical and technical perspective.

Where applicable, discuss:

Privacy

- Does your solution collect or process personal or sensitive information?
- How is user privacy protected?

Fairness and Bias

- Could certain users or groups be disadvantaged?
- Are there potential sources of bias in your dataset or model?

Transparency

- Can users understand or explain how predictions are generated?

Human Oversight

- When should a human review or override AI-generated decisions?

Responsible AI

Explain how your solution supports responsible AI practices within a Canadian context.

Current System Limitations

Discuss the current limitations of your solution, such as:

- Dataset limitations
- Model accuracy
- Processing speed
- Scalability
- Generalization
- User interface limitations

Recommend realistic improvements that could be implemented before the Final Technical Report.

Support your discussion with examples from your own project whenever possible.

5. Reflection and Future Improvements (0.5 Mark)

Reflect on your development experience.

Briefly discuss:

- The most significant challenge encountered
- Important technical decisions made during development
- Lessons learned from testing and evaluation
- Planned improvements before the Final Technical Report

Submission Checklist

Before submitting, ensure your report includes:

- ☐ Current project progress
- ☐ Performance evaluation using appropriate metrics
- ☐ Performance tables and graphs
- ☐ Confusion Matrix (if applicable)
- ☐ Testing procedures and quality assurance activities
- ☐ Mathematical formulas
- ☐ Mathematical calculations
- ☐ Mathematical interpretation of every evaluation metric
- ☐ Ethical considerations
- ☐ System limitations
- ☐ Reflection and future improvements

Marking Rubric (10 Marks)

Criteria	Marks
Current Project Progress	1
Performance Evaluation and Quality Assessment	3
Mathematical Justification and Interpretation	3
Ethical Considerations and System Limitations	2
Reflection and Future Improvements	1
Total	10

Academic Integrity

This is a **group submission** and must represent the original work of all team members.

Your report must:

- Present original analysis and interpretation based on your own AI/ML project.
- Properly acknowledge all datasets, research papers, software libraries, frameworks, documentation, and other resources used.
- Ensure that all mathematical calculations, performance evaluations, testing results, and ethical discussions accurately reflect your own project.

Use of Generative AI (GenAI)

The acceptable use of Generative AI has been discussed during class and communicated through SLATE announcements.

You may use GenAI tools (e.g., ChatGPT, Gemini, Claude, or Copilot) **only to support brainstorming, planning, or idea generation.**

You **must not** copy or submit AI-generated:

- Report text
- Mathematical calculations
- Mathematical explanations
- Python or other programming code
- Figures or tables
- Graphs
- Analyses
- Interpretations
- Conclusions

All submitted work must be calculated, interpreted, and verified by your team.

All submissions will be reviewed using **CopyLeaks** through SLATE. Students can view their AI detection score before final submission and are expected to revise their work if necessary.

Target acceptable GenAI usage for this deliverable: less than 10%.

Academic Misconduct

- Any form of plagiarism or submission of AI-generated text, mathematical calculations, code, figures, tables, graphs, or other content as your own work is strictly prohibited.
- **Any group that submits copied or AI-generated content without proper original work will receive a grade of zero (0) for this deliverable.**
- In addition, violations will be reported in accordance with **Sheridan College's Academic Integrity Policy** and may result in further disciplinary action.
- **Important:** A report that contains evaluation scores without supporting mathematical formulas, calculations, and interpretation does **not** satisfy the requirements of this assessment and will lose marks under the **Mathematical Justification** criterion.